1) a)

$r\cos\left(\theta-\frac{\pi}{3}\right)=1$

$r\cos\theta\cos\frac{\pi}{3}+r\sin\theta\sin\frac{\pi}{3}=1$

$x\cos\frac{\pi}{3}+y\sin\frac{\pi}{3}=1$

$\frac{1}{2}x+\frac{\sqrt{3}}{2}y=1$

$x+\sqrt{3}y=2$

1) b)

$r\cos\left(\theta+\frac{\pi}{6}\right)=1$

$r\cos\theta\cos\frac{\pi}{6}-r\sin\theta\sin\frac{\pi}{6}=1$

$x\cos\frac{\pi}{6}-y\sin\frac{\pi}{6}=1$

$\frac{\sqrt{3}}{2}x-\frac{1}{2}y=1$

$\sqrt{3}x-y=2$

1) c)

$r\sin\left(\theta+\frac{\pi}{4}\right)=1$

$r\sin\theta\cos\frac{\pi}{4}+r\cos\theta\sin\frac{\pi}{4}=1$

$y\cos\frac{\pi}{4}+x\sin\frac{\pi}{4}=1$

$\frac{\sqrt{2}}{2}y+\frac{\sqrt{2}}{2}x=1$

$\sqrt{2}x+\sqrt{2}y=2$

1) d)

$r\sin\left(\theta-\frac{\pi}{6}\right)=4$

$r\sin\theta\cos\frac{\pi}{6}-r\cos\theta\sin\frac{\pi}{6}=4$

$y\cos\frac{\pi}{6}-x\sin\frac{\pi}{6}=4$

$\frac{\sqrt{3}}{2}y-\frac{1}{2}x=4$

$x-\sqrt{3}y=-8$

2) a)

$r = 3 + \cos 2\theta$

$r = 3 + \cos^2\theta - \sin^2\theta$

$r^3 = 3r^2 + r^2\cos^2\theta - r^2\sin^2\theta$

$(x^2 + y^2)^{\frac{3}{2}} = 3x^2 + 3y^2 + x^2 - y^2$

$(x^2 + y^2)^{\frac{3}{2}} = 4x^2 + 2y^2$

2) b)

$r^2 = \sin 2\theta$

$r^2 = 2\sin\theta\cos\theta$

$r^4 = 2r^2\sin\theta\cos\theta$

$(x^2 + y^2)^2 = 2xy$

2) c)

$r^2 = 3\sec 2\theta$

$r^2\cos 2\theta = 3$

$r^2(\cos^2\theta - \sin^2\theta) = 3$

$r^2\cos^2\theta - r^2\sin^2\theta = 3$

$x^2 - y^2 = 3$

2) d)

$r^2 = 9\,\mathrm{cosec}\,2\theta$

$r^2\sin 2\theta = 9$

$r^2(2\sin\theta\cos\theta) = 9$

$2r\sin\theta\, r\cos\theta = 9$

$xy = \frac{9}{2}$

2) e)

$r = 6\cot\theta\,\mathrm{cosec}\,\theta$

$r = 6 \times \frac{\cos\theta}{\sin\theta} \times \frac{1}{\sin\theta}$

$r\sin^2\theta = 6\cos\theta$

$r^2\sin^2\theta = 6r\cos\theta$

$y^2 = 6x$

3)

$(x^2 + y^2)^2 = x^2 - y^2$

$r^4 = r^2\cos^2\theta - r^2\sin^2\theta$

$r^2 = \cos 2\theta$

$\therefore \mathrm{f}(\theta) = \cos 2\theta$

4)

$y = \frac{1}{x}$

$r\sin\theta = \frac{1}{r\cos\theta}$

$r^2 = \frac{1}{\sin\theta\cos\theta}$

$r^2 = \frac{2}{2\sin\theta\cos\theta}$

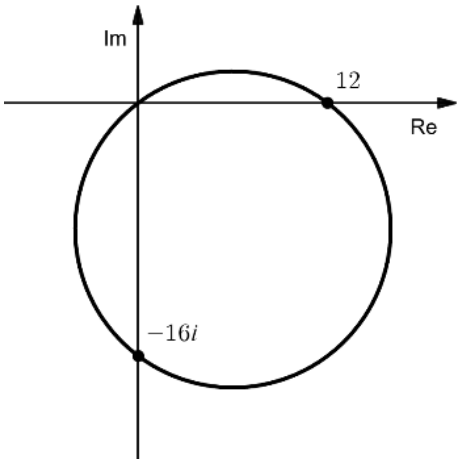
$r^2 = \frac{2}{\sin 2\theta}$

$r^2 = 2\operatorname{cosec} 2\theta$

$\therefore \mathrm{f}(\theta) = 2\operatorname{cosec} 2\theta$

5) a)

Circle centred at $(6, -8)$ with radius 10



5) b)

$(x-6)^2+(y+8)^2=100$

$x^2-12x+36+y^2+16y+64=100$

$x^2-12x+y^2+16y=0$

$r^2\cos^2\theta-12r\cos\theta+r^2\sin^2\theta+16r\sin\theta=0$

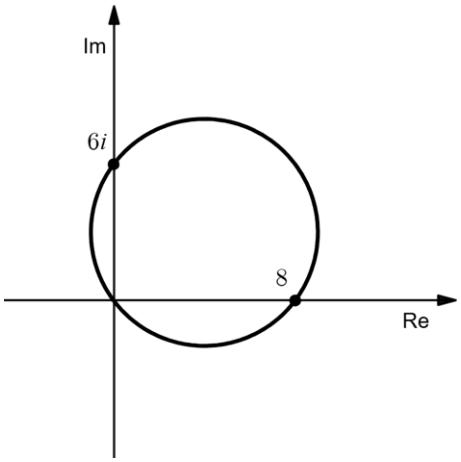
$r^2(\cos^2\theta+\sin^2\theta)-12r\cos\theta+16r\sin\theta=0$

$r^2=12r\cos\theta-16r\sin\theta$

$r=12\cos\theta-16\sin\theta$ as req.

6) a)

Circle centred at $(4,3)$ with radius 5



6) b)

$(x-4)^2+(y-3)^2=25$

$x^2-8x+16+y^2-6y+9=25$

$x^2-8x+y^2-6y=0$

$r^2\cos^2\theta-8r\cos\theta+r^2\sin^2\theta-6r\sin\theta=0$

$r^2(\cos^2\theta+\sin^2\theta)-8r\cos\theta-6r\sin\theta=0$

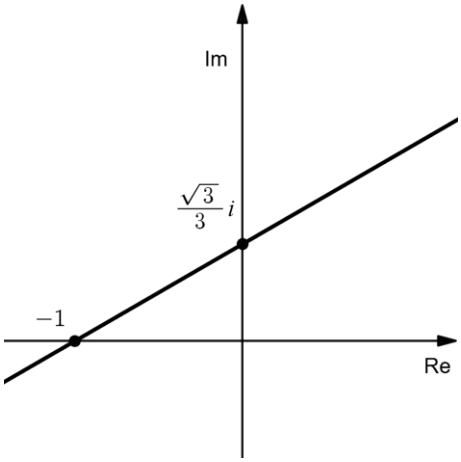
$r^2=8r\cos\theta+6r\sin\theta$

$r=8\cos\theta+6\sin\theta$ as req.

7) a)

$(1,0)$ and $(0,\sqrt{3})$ has midpoint $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ and perpedicular gradient is $\frac{\sqrt{3}}{3}$

$y-\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{3}\left(x-\frac{1}{2}\right)\Rightarrow y=\frac{\sqrt{3}}{3}x-\frac{\sqrt{3}}{6}+\frac{\sqrt{3}}{2}\therefore y=\frac{\sqrt{3}}{3}x+\frac{\sqrt{3}}{3}$



7) b)

$y=\frac{\sqrt{3}}{3}x+\frac{\sqrt{3}}{3}$

$r\sin\theta=\frac{\sqrt{3}}{3}\,r\cos\theta+\frac{\sqrt{3}}{3}$

$3r\sin\theta-\sqrt{3}\,r\cos\theta=\sqrt{3}$

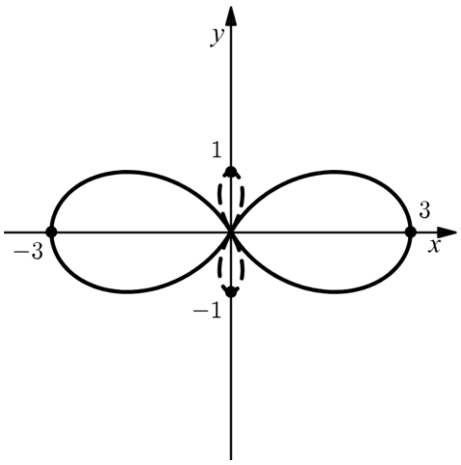
$\sin(\theta-\alpha)=\sin\theta\cos\alpha-\cos\theta\sin\alpha$

$\cos\alpha=3$ and $\sin\alpha=\sqrt{3}\Rightarrow R=\sqrt{12}$ and $\alpha=\arctan\left(\frac{\sqrt{3}}{3}\right)=\frac{\pi}{6}$

So that $r\sin\theta-\sqrt{3}\,r\cos\theta=\sqrt{3}$ becomes $\sqrt{12}r\sin\left(\theta-\frac{\pi}{6}\right)=\sqrt{3}$

$\therefore r=\frac{1}{2}\text{cosec}\left(\theta-\frac{\pi}{6}\right)$ as req.

8) a)



8) b)

$1 + 2\cos 2\theta = \sqrt{3}\sec\theta$

$1 + 2(2\cos^2\theta - 1) = \sqrt{3}\sec\theta$

$1 + 4\cos^2\theta - 2 = \sqrt{3}\sec\theta$

$4\cos^2\theta - 1 = \sqrt{3}\sec\theta$

$4\cos^3\theta - \cos\theta = \sqrt{3}$

$4\cos^3\theta - \cos\theta - \sqrt{3} = 0$

9)

$\cosh^2 x - \sinh^2 x = 1 \Rightarrow 1 + \sinh^2 x = \cosh^2 x$

So that $r = \sqrt{1 + \sinh^2\theta} = \sqrt{\cosh^2\theta} = \cosh\theta$ since $r > 0$

For $y = x$ then $r\sin\theta = r\cos\theta$

So at the points of intersection: $\cosh\theta\sin\theta = \cosh\theta\cos\theta$

Since $\cosh\theta \neq 0$, then solve $\sin\theta = \cos\theta$

$\Rightarrow \tan\theta = 1 \therefore \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \ldots$

For $\theta = \frac{\pi}{4}$:

$r = \cosh\frac{\pi}{4} \Rightarrow x = r\cos\theta = \cosh\frac{\pi}{4}\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}$ and

$y = r\sin\theta = \cosh\frac{\pi}{4}\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}$

So: $\left(\frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}, \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}\right)$

For $\theta = \frac{5\pi}{4}$:

$r = \cosh\frac{5\pi}{4} \Rightarrow x = r\cos\theta = \cosh\frac{5\pi}{4}\cos\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}$ and

$y = r\sin\theta = \cosh\frac{5\pi}{4}\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}$

So: $\left(-\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}, -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}\right)$

$\therefore \left(\frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}, \frac{\sqrt{2}}{2}\cosh\frac{\pi}{4}\right)$ and $\left(-\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}, -\frac{\sqrt{2}}{2}\cosh\frac{5\pi}{4}\right)$

Or $\left(\frac{\sqrt{2}}{4}\left(e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}\right), \frac{\sqrt{2}}{4}\left(e^{\frac{\pi}{4}} + e^{-\frac{\pi}{4}}\right)\right)$ and $\left(-\frac{\sqrt{2}}{4}\left(e^{\frac{5\pi}{4}} + e^{-\frac{5\pi}{4}}\right), -\frac{\sqrt{2}}{4}\left(e^{\frac{5\pi}{4}} + e^{-\frac{5\pi}{4}}\right)\right)$

10) a)

$x^3 - y^3 = 3xy$

$r^3\cos^3\theta - r^3\sin^3\theta = 3r^2\cos\theta\sin\theta$

$r^3(\cos^3\theta - \sin^3\theta) = 3r^2\cos\theta\sin\theta$

Since $r > 0$ then $\mathrm{f}(\theta) = \frac{3\cos\theta\sin\theta}{\cos^3\theta - \sin^3\theta}$

10) b)

Since $\cos\left(\frac{3\pi}{2} - \theta\right) = \cos\frac{3\pi}{2}\cos\theta + \sin\frac{3\pi}{2}\sin\theta = -\sin\theta$

and $\sin\left(\frac{3\pi}{2} - \theta\right) = \sin\frac{3\pi}{2}\cos\theta - \cos\frac{3\pi}{2}\sin\theta = -\cos\theta$ then:

$$\mathrm{f}\left(\frac{3\pi}{2} - \theta\right) = \frac{3\cos\left(\frac{3\pi}{2} - \theta\right)\sin\left(\frac{3\pi}{2} - \theta\right)}{\cos^3\left(\frac{3\pi}{2} - \theta\right) - \sin^3\left(\frac{3\pi}{2} - \theta\right)}$$

$$= \frac{3(-\sin\theta)(-\cos\theta)}{(-\sin\theta)^3 - (-\cos\theta)^3} = \frac{3\cos\theta\sin\theta}{-\sin^3\theta - -\cos^3\theta} = \frac{3\cos\theta\sin\theta}{\cos^3\theta - \sin^3\theta}$$

10) c)

Since $\cos\left(\frac{3\pi}{2} - \theta\right) = -\sin\theta$ and $\sin\left(\frac{3\pi}{2} - \theta\right) = -\cos\theta$ this represents $(x, y) \mapsto (-y, -x)$ which is a reflection in the line $y = -x$

So the line of symmetry is $y = -x$

10) d)

When $\theta = \frac{\pi}{2}$, $\mathrm{f}\left(\frac{\pi}{2}\right) = \frac{3\cos\frac{\pi}{2}\sin\frac{\pi}{2}}{\cos^3\frac{\pi}{2} - \sin^3\frac{\pi}{2}} = 0$

When $\theta = \pi$, $\mathrm{f}(\pi) = \frac{3\cos\pi\sin\pi}{\cos^3\pi - \sin^3\pi} = 0$

So for $\frac{\pi}{2} \leq \theta \leq \pi$ it is symmetrical across $y = -x$ and starts and ends at the origin therefore it must be a loop.

11) a)

$13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta$

$= 13\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7\sin^2\theta$

$= 13\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7(1 - \cos^2\theta)$

$= 6\cos^2\theta - 3\sqrt{3}\sin 2\theta + 7$

$= 6\left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right) - 3\sqrt{3}\sin 2\theta + 7$

$= 3 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta + 7$

$= 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta$ as req.

11) b)

$13x^2 - 6\sqrt{3}xy + 7y^2 = 1$

$13r^2\cos^2\theta - 6\sqrt{3}r^2\cos\theta\sin\theta + 7r^2\sin^2\theta = 1$

$r^2\left(13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta\right) = 1$

$r^2 = \frac{1}{13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta}$

Since $13\cos^2\theta - 6\sqrt{3}\cos\theta\sin\theta + 7\sin^2\theta = 10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta$ then

$r^2 = \frac{1}{10 + 3\cos 2\theta - 3\sqrt{3}\sin 2\theta}$

Consider $3\cos 2\theta - 3\sqrt{3}\sin 2\theta$ and so

$R\sin(2\theta + \alpha) = R\sin 2\theta\cos\alpha + R\cos 2\theta\sin\alpha$

$R = \sqrt{3^2 + \left(3\sqrt{3}\right)^2} = 6$ and $\alpha = \arctan\left(\frac{3}{3\sqrt{3}}\right) = \frac{\pi}{6}$

So that

$r^2 = \frac{1}{10 - 6\sin\left(2\theta - \frac{\pi}{6}\right)}$

$\therefore$ max and min of r^2 are $r^2 = \frac{1}{16}$ and $r^2 = \frac{1}{4}$

So the greatest distance is $\frac{1}{2}$ and the least distance is $\frac{1}{4}$